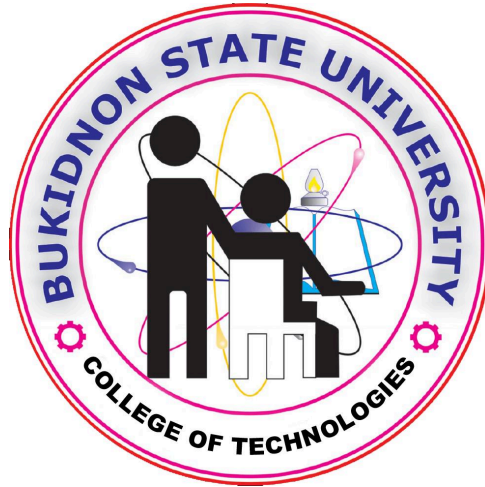


BUKIDNON STATE UNIVERSITY
COLLEGE OF TECHNOLOGIES
INFORMATION TECHNOLOGY DEPARTMENT



**Title: Building a Robotic Arm Using Potentiometers,
Servos, and Arduino**

A Project Presented to
The Information Technology Department
In Partial Fulfillment of the Requirement for the Course
T194/IT 124 - Principle of Logic Design

by
Joenil Acero
Rayden Ivan Delfin
Irijah Maxim Magdua
Queen de los Reyes

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Realino Indapan, III
Instructor

Introduction

Building a robotic arm with potentiometers, servos, and Arduino has been an exciting and educational experience for our group. This project allowed us to explore robotics and electronics while working together to design and construct a robotic arm that mimics human movements. We gained hands-on experience in programming the Arduino to control the arm and learned how to effectively integrate sensors and actuators. This collaboration enhanced our technical skills and inspired our passion for technology and problem-solving.

Objectives

The primary objectives of our robotic arm project are:

Design and Construct: To collaboratively design and build a robotic arm that mimics human-like movement using servo motors and potentiometers.

Programming: To program the Arduino microcontroller to effectively control the arm's movements.

Integration: To understand the integration of sensors (potentiometers) and actuators (servos) in a robotic system.

Team Collaboration: To enhance teamwork and problem-solving skills through collaborative efforts.

Design and Methodology

Arm's Movement Definition

The robotic arm consists of four joints, each providing a degree of freedom (DOF):

Base Rotation: Controlled by servo1.

Shoulder Joint: Controlled by servo2.

Elbow Joint: Controlled by servo3.

Gripper: Controlled by servo4.

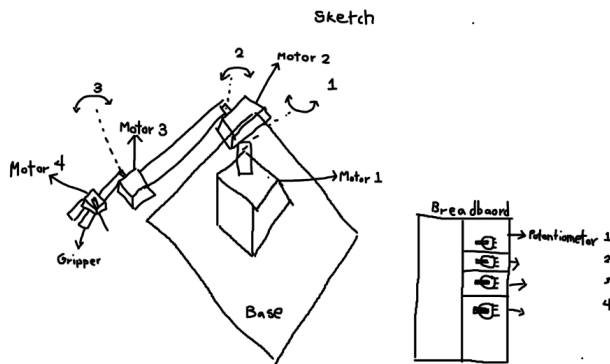


Figure 1: Sketch

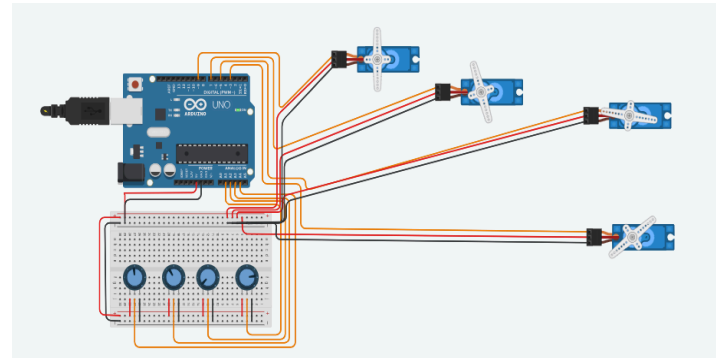


Figure 2: Flowchart

Challenges and Solutions

Challenges Faced

During the development of our robotic arm, we encountered several significant challenges:

Design Compatibility: One of the primary difficulties was creating a design that utilized compatible materials for the servo motors. Ensuring that the materials could withstand the operational demands while maintaining functionality was crucial.

Potentiometer Reliability: We experienced intermittent issues with the potentiometers, which occasionally failed to function as expected. This inconsistency was likely due to fluctuations in the current flow necessary for controlling the motors effectively.

Solutions Implemented

To address these challenges, we adopted the following solutions:

Material Selection: We opted for lightweight materials, specifically plastic, to construct the arm. This choice not only reduced the overall weight of the robotic arm but also ensured that the servo motors could handle the load without strain. Additionally, the use of lightweight materials facilitated smoother movements of the arm.

Improved Potentiometer Functionality: To enhance the reliability of the potentiometers, we carefully reviewed and optimized the electrical connections. By ensuring a stable flow of current, we minimized the chances of malfunction, allowing for more consistent and accurate control of the arm's movements.

Through these solutions, we were able to create a more effective and reliable robotic arm, enabling us to achieve our project goals while enhancing our understanding of material properties and electrical systems in robotics.

Conclusion

In conclusion, our robotic arm project taught us a lot about design, teamwork, and problem-solving. By overcoming challenges like material compatibility and potentiometer reliability, we learned the importance of choosing the right materials and ensuring proper electrical connections. These experiences not only helped us improve our technical skills but also strengthened our collaboration as a team.

Overall, this project was a rewarding journey that sparked our interest in robotics and engineering. We are proud of what we accomplished together and excited about the possibilities for future projects. This experience has inspired us to continue exploring technology and its applications in creative ways.

Source Code:

```
1  #include <Servo.h>
2
3  // Define the servo objects
4  Servo servo1; // Servo for joint 1 for base
5  Servo servo2; // Servo for joint 2
6  Servo servo3; // Servo for joint 3
7  Servo servo4; // Servo for joint 4 gripper/claw
8
9  // Define potentiometer pins
10 const int potPin1 = A1; // Potentiometer for servo 1 BASE
11 const int potPin2 = A2; // Potentiometer for servo 2 FIRST JOINT
12 const int potPin3 = A3; // Potentiometer for servo 3 SECOND JOINT
13 const int potPin4 = A4; // Potentiometer for servo 4 GRIPPER
14
15 void setup() {
16     // Attach the servos to the PWM pins
17     servo1.attach(3); // Attach servo 1 to pin 3
18     servo2.attach(5); // Attach servo 2 to pin 5
19     servo3.attach(7); // Attach servo 3 to pin 7
20     servo4.attach(9); // Attach servo 4 to pin 9
21 }
22
23 void loop() {
24     // Read values from the potentiometers
25     int potValue1 = analogRead(potPin1); // Read pot value for servo 1
26     int potValue2 = analogRead(potPin2); // Read pot value for servo 2
27     int potValue3 = analogRead(potPin3); // Read pot value for servo 3
28     int potValue4 = analogRead(potPin4); // Read pot value for servo 4
29
30     // Scale the potentiometer values from 0-1023 to 0-180
31     int servoPos1 = map(potValue1, 0, 1023, 0, 180);
32     int servoPos2 = map(potValue2, 0, 1023, 0, 180);
33     int servoPos3 = map(potValue3, 0, 1023, 0, 180);
34     int servoPos4 = map(potValue4, 0, 1023, 0, 180);
35
36     // Set the servo positions
37     servo1.write(servoPos1);
38     servo2.write(servoPos2);
39     servo3.write(servoPos3);
40     servo4.write(servoPos4);
41
42     // Small delay to stabilize the readings
43     delay(15);
44 }
45 }
```

Figure 3: Code Snippet

